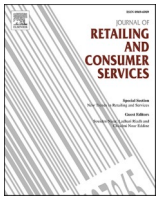
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Navigating trust and the personalization-privacy paradox in AI-driven e-commerce: A consumer pathway framework

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ABSTRACT

As brands begin integrating artificial intelligence (AI) into their e-commerce systems, understanding consumers' navigation of trust, privacy, and personalization within this domain will equip brands with successful consumer-centered, strategic approaches. This study explores how consumers calibrate trust in AI-enabled e-commerce, with particular attention to the personalization-privacy paradox, which is the tension between desiring tailored recommendations and resisting the data practices required to produce them. Through qualitative interviews and thematic analysis, the researchers established four interlocking mechanisms: affective orientations, stereotypical cognition, trust aversion, and trust calibration. These mechanisms were synthesized into a novel four-pathway framework called UCCG—utility-led (U), control-led (C), communication-led (C), and governance-led (G)—that explained how patterned combinations of trust and privacy perceptions generated distinct routes to commitment, reuse, and loyalty in AI-enabled e-commerce platforms. The findings demonstrated that consumer trust in AI is not a linear or uniform process, but a continuously adjusted, experience-based judgment shaped by the personalization-privacy paradox and by factors, such as AI technical performance, brands' transparency practices, and perceived data governance practices. Transparency and user control emerged as critical trust antecedents, though their effectiveness varied across consumer pathways. This study extends personalization-privacy paradox and commitment-trust theory into the e-commerce AI context by theorizing trust and privacy tension as a dynamic, calibrated process, and offers managerial guidance for brands designing trust-centered e-commerce AI experiences.

1. Introduction

Artificial intelligence (AI) has moved from experimental pilots to embedded infrastructure across customer-facing touchpoints, reshaping the customer journey from product discovery to post-purchase service (Deloitte, 2026; Kukreti and Yadav, 2026; Rapoport et al., 2025; P. Wang et al., 2024; Zheng and Li, 2024). In marketing, AI is defined as the use of computational machinery to replicate human capabilities, including performing physical tasks, thinking, and experiencing emotions (M.-H. Huang and Rust, 2021; Syam and Sharma, 2018). The pace of AI adoption is accelerating: by 2028, approximately 33% of e-commerce enterprises are projected to adopt agentic AI, and the AI-driven e-commerce market is expected to reach approximately $22.6 billion by 2032 (SellersCommerce, 2025). As AI becomes more integrated into everyday shopping, understanding how consumers navigate these AI-enabled interfaces has become critical for retailing scholarship and practice.

Among all AI-powered applications in marketing, e-commerce AI is a particularly pivotal area. It encompasses systems and algorithms that facilitate online purchasing, generate personalized product recommendations, and support tasks such as feature comparison, payment processing, and personalized marketing messages (Jayakumar et al., 2024; Kumar et al., 2024; Teodorescu et al., 2023), which collectively contribute to a faster and more convenient online shopping experience. Amazon's generative AI (GenAI) shopping assistant, Rufus, exemplifies these capabilities: it generates review summaries and answers consumer questions to help shoppers find products that meet their needs (Amazon, 2024). Such applications have begun to reshape how consumers perceive and evaluate brands, as well as purchase products, shifting parts of the shopping experience from active search to AI-mediated

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guidance. As firms embed AI agents into shopping journeys, design choices around personalization, transparency, and interaction style increasingly shape consumer experience and downstream buying behavior (Farooq et al., 2024). These shifts raise important questions about how consumers respond to AI integration, particularly when it intersects with trust and privacy concerns.

Trust has emerged as a central mechanism through which consumers translate satisfaction with AI-driven personalization into continued use and loyalty (Bunea et al., 2024; Ding and Najaf, 2024; McLean et al., 2021; Soleimani, 2022). Research on trust in AI shopping has largely followed two parallel paths. One path examines how AI features shape consumer trust, including anthropomorphic interfaces (Calhoun et al., 2019; Ding and Najaf, 2024; Igarashi et al., 2026), perceived algorithmic transparency (Ma et al., 2026) and user familiarity (Vu et al., 2026), and the effects of AI disclosure on purchase behavior (T.-G. Li et al., 2024; Luo et al., 2019). A second path examines privacy concerns as a force that can affect trust and shape whether consumers are willing to engage with personalized services at all (Aguirre et al., 2015; Awad and Krishnan, 2006; Karwatzki et al., 2017). Nevertheless, this second path has primarily treated privacy as a standalone tradeoff between data disclosure and personalization benefits and has not integrated its findings into the commitment-trust process that explains how trust translates into commitment and loyalty (Morgan and Hunt, 1994).

This privacy-versus-personalization tension is especially pressing in AI-enabled e-commerce. Personalization depends on continuous behavioral data collection and inference, which intensifies perceived privacy risks (Canhoto et al., 2024; Ma et al., 2026). Survey evidence indicates that approximately 82% of consumers are concerned that AI may compromise their online privacy (The Customer Data Platform Resource, 2024), and these concerns are amplified by persistent implementation risks, such as algorithmic bias and opaque governance (Elliott and Soifer, 2022; Kumar et al., 2024; Manyika et al., 2019). While some consumers are willing to share their personal data when shopping online, many others hesitate due to uncertainty about illegal data mining (Madanchian, 2024; Wang et al., 2021). The result is the personalization-privacy paradox: consumers desire tailored recommendations while simultaneously resisting the data practices required to produce them (Awad and Krishnan, 2006; Cloarec, 2020). Trust and the personalization-privacy paradox play a crucial role in e-commerce AI and cannot be understood in isolation from each other, yet existing work has largely examined trust and privacy as separable, linear predictors of adoption rather than as dynamically interlinked forces that shape distinct consumer orientations (Aguirre et al., 2015; Ameen et al., 2022; Ma et al., 2026; Talebi et al., 2025).

Despite the growing evidence that trust in AI is neither linear nor uniform (Gu et al., 2024; Luo et al., 2019), little is known about how consumers form and adjust trust in AI recommendations in everyday shopping across the tension between personalization and privacy, and how these trust calibration processes unfold differently across consumer types. This study addresses that gap by introducing four consumer pathways, derived from qualitative interviews and grounded in the integration of commitment-trust theory (Morgan and Hunt, 1994) and the personalization-privacy paradox (Awad and Krishnan, 2006; Cloarec, 2020). At the center of the framework is trust calibration: the consumer-driven, multi-dimensional process through which trust in e-commerce AI is actively formed, adjusted, or discontinued in response to companies' privacy practices, such as transparency and the level of user control they provide, rather than assumed as a stable antecedent of adoption. The pathways explain how patterned combinations of trust and privacy perceptions generate distinct routes to commitment and loyalty in AI-enabled e-commerce.

2. Literature review

2.1. Personalization-privacy paradox

The personalization-privacy paradox, originating from Awad and Krishnan's (2006) study of online profiling, captures a foundational tension in digital consumer behavior: consumers who most value information transparency are simultaneously the least willing to be profiled for personalization. Since its formulation, the paradox has been applied across multiple retail and consumer service contexts: online advertising effectiveness (Aguirre et al., 2015), smart retail environments and shopping mall loyalty (Ameen et al., 2022), physical retail AI personalization (Canhoto et al., 2024), AI personalization and the roles of privacy in advertising (Qadri et al., 2026), and AI-driven recommendation effects (Lo et al., 2026). A consistent insight has emerged across some studies that privacy concerns do not only coexist with trust but also actively shape whether consumers will engage with personalized offerings at all (Connolly and Bannister, 2007; Qadri et al., 2026; Shukla, 2014). The paradox also implies a trust hurdle: even with information transparency, privacy-sensitive consumers may remain unwilling to be profiled and may never engage deeply enough to benefit from personalization (Awad and Krishnan, 2006), highlighting the need for trust-building strategies that go beyond disclosure alone. Despite the paradox's growing relevance in AI-enabled retailing, prior applications have largely treated it as a static cost-benefit calculation, rarely exploring how trust is built and sustained through relational exchange.

The empirical evidence reveals that the same paradox produces different consumer responses depending on how trust is established. Aguirre et al. (2015), for example, showed that covert data collection evoked vulnerability and diminished consumer engagement, whereas overt disclosure paired with credible trust cues (e.g., third-party endorsements) increased receptivity to personalized advertising. More recent AI-enabled retailing research has complicated this picture: in some contexts, consumers reported relatively low privacy concern in response to personalization, viewing data sharing as a routine condition of digital service use (Ameen et al., 2022; Canhoto et al., 2024). These findings are not contradictory but illuminate a crucial pattern. The paradox does not operate uniformly; some consumers accept data exchange as ordinary digital practice while others resist it even when personalization benefits are evident. This variability raises a question that existing research has not addressed: whether consumers navigate the paradox through systematically different orientations, and if so, how these orientations shape trust and commitment in AI-enabled e-commerce. Addressing this question requires examining the paradox alongside a relational framework that explains how trust translates into loyalty—namely, the commitment-trust theory.

2.2. Commitment-trust theory in AI

Trust has long been theorized as the foundation of relational exchange. Morgan and Hunt's (1994) commitment-trust theory shows that trust (i.e., "confidence in an exchange partner's reliability and integrity") and commitment (i.e., "an enduring desire to maintain a valued relationship") are the central mediating variables in successful relational exchange (p. 23). Within this framework, factors such as communication, shared values, relationship benefits, termination costs, and perceived opportunistic behavior influence relational outcomes, but only through their mediating effects on trust and commitment (Morgan and Hunt, 1994; Ozuem et al., 2023). Applied to e-commerce AI contexts, this theory helps explain why trust in AI tools influences both behavioral intentions and longer-term consumer-brand relationships (Alagarsamy and Mehrolia, 2023; Teodorescu et al., 2023). A growing body of work in retailing and consumer services has examined how various design choices influence consumers' trust in AI-enabled retail interfaces, including language style (M. Li and Wang, 2023), informational and emotional support for Generation Z (Gen Z; Guo and Luo,

2023), chatbot identity (Lu and Zhang, 2025), affective attachment (Bilal et al., 2024), and anthropomorphic chatbot features (Igarashi et al., 2026). Nonetheless, across these studies, trust is mainly treated as a relatively stable factor that AI design either does or does not produce, rather than as a dynamically calibrated process that interacts with privacy concerns, leaving unexplained when and for whom trust forms or fails.

Consumer trust in AI does not follow a linear pathway as the same AI design feature can build trust in one context and erode it in another, and AI disclosure illustrates this most clearly. On the one hand, when consumers learn how an AI's recommendation logic works, they often trust the system more (Oyekunle et al., 2024). On the other hand, when a chatbot's identity is disclosed before an interaction began, purchase rates can drop sharply even when the AI performed as well as a human agent (Luo et al., 2019). A similar duality appears with anthropomorphism, AI-generated content, and chatbot service failures: each can foster engagement under some conditions but trigger resistance when it violates consumer expectations about authenticity, control, or competence (Brüns and Meißner, 2024; Gu et al., 2024; Y.-S. (Sandy) Huang and Dootson, 2022; Sands et al., 2024). Beyond individual interactions, broader cultural narratives about AI failures and identity-related concerns when brands use AI-generated models further constrain consumer trust in ways that brands cannot fully control (Gursoy et al., 2019; Oyekunle et al., 2024; Sands et al., 2024). These findings suggest that trust in e-commerce AI does not function as a static consumer attribute. Instead, it might form or fail dynamically depending on various components, such as disclosure conditions, AI performance, the data practices that accompany personalization, and more.

What emerges from this literature is a theoretical disconnect. Table 1 synthesizes critical studies on trust and privacy in AI-driven online shopping. Reading across these studies, three patterns emerge that motivate the present research. First, the theoretical frameworks used (i.e., privacy calculus, trust transfer, computers-as-social-actors, attribution theory, social identity, and others) largely operate in isolation, and few studies attempt to integrate trust-based and privacy-based explanations of consumer behavior. Second, the methodologies are dominated by surveys and controlled experiments, leaving limited space for qualitative work that captures how consumers themselves articulate their reasoning about AI, trust, and privacy. Third, the same design features (e.g., disclosure, anthropomorphism, personalization) repeatedly produce divergent consumer responses across studies, pointing to underlying consumer heterogeneity that existing frameworks have not theorized. Together, these patterns indicate the need for an integrative, qualitative investigation of how trust and privacy operate jointly across different consumer orientations to generate distinct pathways to commitment and loyalty—the contribution this study aims to make.

3. Methodology and methods

This study was conducted as an exploratory research endeavor aimed at examining consumers' perspectives on privacy and trust within the AI-enabled e-commerce landscape. The research team used a purposive, criterion-based approach (Morse and Clark, 2019; Stalmeijer et al., 2024), establishing two inclusion criteria: a minimum age of 18 and residence in the United States (U.S.). Recruitment was conducted through multiple channels, including snowball referrals from initial participants, university student networks, and the research team's personal and professional networks. Importantly, prior experience with AI tools was not required for participation. Participants, therefore, reflected a range of familiarity with AI-enabled shopping, from those who were unaware they had interacted with AI in e-commerce to those who often used tools such as Amazon's Rufus or ChatGPT. This variability is analytically productive because it captures consumers at different points along the trust calibration process, rather than sampling only those who have already formed settled attitudes toward AI. This approach aligns with criterion sampling, a form of purposeful sampling in which all cases meeting predetermined criteria of relevance are selected for study (Morse and Clark, 2019; Patton, 2015). The sampling strategy prioritized theoretical relevance over demographic representativeness. This emphasis on information richness aligns with the primary objective of qualitative inquiry, which focuses on conceptual transferability rather than statistical generalizability (Lincoln and Guba, 1985; Patton, 2015; Stalmeijer et al., 2024).

The sample consists of 31 participants with an average of 2.06 years of AI usage experience from various U.S. states. The Northeast region has the highest representation, followed by the Midwest and the Southern regions. The sample also encompassed a range of occupations and employment statuses, including undergraduate and graduate students (22/31), as well as full-time and part-time individuals employed in diverse roles across healthcare, finance, marketing, communication, and administration. The sample is concentrated among 18-24-year-old individuals (i.e., 26 of 31 participants; see Table 2). This age concentration is theoretically appropriate for three reasons. First, younger consumers, particularly Gen Z, are the most active adopters and users of AI-enabled shopping tools, making them an information-rich population for exploring trust and privacy perceptions in this domain. Specifically, industry data in 2025 showed that Gen Z consumers were nearly three times more likely than Gen X to use AI for product research (Commerce, 2025), and nearly 43% of Gen Z used AI tools for the holiday shopping season (Deloitte, 2025). Critically, this generation simultaneously adopts and distrusts AI shopping tools: while 61% of Gen Z consumers used AI for purchases (EMARKETER Editors, 2025; PayPal, 2025), only approximately 23% of this population trusted AI recommendations more than human ones (Commerce, 2025). This paradox of high adoption alongside low trust makes Gen Z a particularly productive population for investigating the personalization-privacy paradox and trust calibration processes that this study examines.

Second, this sampling approach aligns with published research in retailing and consumer services that has focused on younger consumers when investigating AI-related phenomena, recognizing that this demographic's behaviors and perceptions often signal emergent patterns that subsequently diffuse to broader consumer segments (Bunea et al., 2024; Guo and Luo, 2023). Third, consistent with Lincoln and Guba's (1985) framework for trustworthiness in qualitative research, the findings should be assessed in terms of conceptual transferability rather than statistical generalizability. This study's contribution lies in identifying theoretical mechanisms and consumer pathways that offer conceptual insight into how trust and privacy operate in AI-enabled e-commerce; whether these mechanisms hold across other consumer segments remains a productive direction for future research. Detailed contextual descriptions are provided throughout the manuscript to enable readers to assess the applicability of these findings to other populations and settings (Lincoln and Guba, 1985; Stalmeijer et al., 2024).

The four researchers in this study come from different disciplinary backgrounds and countries (the U.S. and the United Kingdom). Each researcher played a different role, including facilitating interviews, engaging with participants, designing questions, analyzing data, and proofreading. The implications of this multi-researcher, multi-national composition for analytical rigor are discussed in Section 4 Data Analysis. To address the study's objectives, the research team members came up with 19 exploratory open-ended questions, allowing participants to elaborate on their perceptions of AI integration and usage in e-commerce (see Appendix A). Therefore, participants had unlimited freedom to express their views toward each question. In the interviews, the use of AI in e-commerce was clearly explained as any platform that employed algorithms or statistical models to provide recommendations similar to human capabilities, such as performing thinking or experiencing emotions (e.g., AI chatbots on e-commerce platforms or GenAI for product recommendations) (M.-H. Huang and Rust, 2021; Longoni et al., 2019; Syam and Sharma, 2018). The interviews, which lasted around 30-40 min, were conducted via Zoom, with respondents being asked multiple open-ended and yes/no questions with probes (see Appendix

Table 1
Previous peer-reviewed studies on trust and privacy in AI-driven shopping experience.

Study (Author, year)	Theoretical Framework	Methodology	AI context	Main findings
Awad and Krishnan (2006)	Privacy calculus (cost-benefit of disclosure)	Survey (n > 400)	Online personalization and data profiling	Coined the personalization–privacy paradox. Consumers who desire more information transparency are less willing to be profiled for personalization. Privacy-sensitive users opt out despite transparency measures, posing a dilemma for marketers.
Aguirre et al. (2015)	Trust transfer theory; personalization-privacy paradox	Field study + 3 lab experiments	Personalized online ads (Facebook & web)	Demonstrated paradox: Overt data collection (with consent) led to higher click-through and ad effectiveness with deeper personalization, whereas covert data collection caused feelings of vulnerability and lower response rates. Trust signals (e.g. third-party endorsements) offset the negative effects of privacy concern.
Luo et al. (2019)	Social cognition of AI; trust as mediator of AI acceptance (implicit)	Field experiment (randomized trial)	Sales chatbot vs. human agents (outbound calls in finance)	An undisclosed AI chatbot performed as well as an expert human in converting sales but disclosing the chatbot before the interaction caused a 79.7% drop in purchase rate. Consumers who knew it was a bot found it less knowledgeable and empathetic. Disclosing the AI only after engagement mitigated this effect.
Poushneh (2021)	Social cognition (enactive social interaction); human-computer interaction theory	Two lab experiments	AI voice assistants (mobile and smart speaker: e.g., Amazon Alexa, Microsoft Cortana)	Auditory social interaction with a voice AI (natural, conversational speech) increased consumers' trust in the assistant and positively impacted brand affect. Trusted voice assistants led to higher likelihood of purchase via that assistant. Conversely, unexpected or "surprising" agent behavior undermined perceived control and trust, reducing usage.
Kamoonpuri and Sengar (2023)	Diffusion of Innovations	A multi-step approach combining qualitative and quantitative techniques (i.e., 37 barriers identified through a literature review and 16 exploratory interviews with AI experts and data scientists; Fuzzy Analytic Hierarchy Process; pairwise comparison questionnaires completed by 37 experts in the fields of AI, Natural Language Processing, and Machine Learning)	AI-enabled Virtual Assistants (both text-based chatbots and voice bots) within the Indian retail sector	The lack of control in conversations with AI virtual assistants was ranked as the number one barrier. Psychological discomfort, including feelings of creepiness when interacting with excessively human-like bots and a general lack of trust regarding data privacy, significantly hindered user acceptance. Retailers faced a major challenge in balancing Explainable AI with performance: High-performing models were often black boxes, but transparency was required for trust.
Gu et al. (2024)	Computers as Social Actors (CASA); Attribution theory	Survey (n = 462) with SEM analysis	Customer service chatbot after service failure	After a chatbot service failure, anthropomorphic cues (friendly persona, empathy) boosted consumers' sustained trust in the AI. Users attributed failures to external factors when the bot seemed empathic. AI anxiety negatively moderated this trust recovery.
Lu and Zhang (2025)	Social identity & personification in AI (implied)	Experiments	Chatbot identity design (gender, name, etc.) in retail services	Varied chatbot identities (e.g., different genders/names) and product contexts to examine trust. Found that matching chatbot identity appropriately to the context can enhance consumer trust in the service. Implied that one-size-fits-all bot personas may not maximize trust across contexts.
Ma et al. (2026)	Cognitive-affective personality system (CAPS) theory; Functional trust (i.e., belief in the AI's reliability and data processing) and human-like trust (i.e., emotional connection based on the AI's personification and interaction style)	Field/scenario/online experiments and digital product test	Unidirectional AI Word-of-Mouth (WOM) mimics the static AI summary blocks found at the top of Amazon or JD.com review pages; Interactive AI WOM mimics conversational AI assistants like Amazon Rufus	The study demonstrated a matching effect where consumers preferred unidirectional AI summaries for utilitarian products and interactive AI assistants for hedonic products. This preference was driven by functional trust in the AI's efficiency for practical goods and human-like trust for pleasure-oriented items. However, high privacy concerns acted as a significant barrier, weakening these preferences and reducing the overall willingness to adopt

(*continued on next page*)

Table 1 (*continued*)

Study (Author, year)	Theoretical Framework	Methodology	AI context	Main findings
Igarashi et al. (2026)	Theory of Avatar Marketing, focusing on a 2x2 taxonomy of anthropomorphism: Form Realism (human-like vs. cartoon appearance) and Behavioral Realism (warmth vs. competence response styles); Expectancy-Disconfirmation Theory	Randomized controlled field experiment	Rule-based chatbots as critical customer service and sales touchpoints in the retail industry	AI-generated recommendations regardless of the product type. The main findings suggested that competence-oriented chatbots helped foster initial trust during the exploration phase (one-time orders), while warm, empathetic chatbots were more effective at maintaining that trust and driving loyalty during the maturity phase (subscriptions). Additionally, the authors identified the need to investigate trust and risk perception as key customer characteristics in future research to better understand how anthropomorphic cues influenced decision-making.

Table 2
Sample's demographics.

No.	Gender	Age range	AI usage duration (years)
1	Male	18-24	2
2	Other	18-24	1
3	Female	18-24	1
4	Male	25-34	2
5	Male	25-34	1
6	Female	18-24	2
7	Male	18-24	2
8	Female	35-44	2.5
9	Female	18-24	1
10	Female	25-34	1.5
11	Female	18-24	1
12	Female	18-24	2
13	Female	25-34	3
14	Female	18-24	3
15	Male	18-24	4
16	Female	18-24	2
17	Female	18-24	2
18	Female	18-24	2
19	Female	18-24	1
20	Female	18-24	1
21	Female	18-24	1
22	Female	18-24	2
23	Female	18-24	3
24	Male	18-24	1
25	Female	18-24	3
26	Female	18-24	2
27	Female	18-24	1
28	Male	18-24	3
29	Female	18-24	5
30	Male	18-24	2
31	Female	18-24	4

A). A semi-structured interview format was chosen due to its predefined scope and topics (Ozuem et al., 2023). The time length of the interviews also allowed participants to share enough insights for deep understanding of their perceptions of AI chatbots used for product recommendations (Azemi and Ozuem, 2023). The interview period lasted around 3 months, from July 2025 to October 2025, until data saturation (Glaser and Strauss, 2017; Maxwell, 1992), which was reached at the 31st interview. Additionally, this study adopted the qualitative interview question format from Wang's study (Wang, 2023) but adapted it to align with the privacy-trust theme.

4. Data analysis

The collected data were analyzed using thematic analysis to enhance the conceptualization of empirical findings (Gioia et al., 2013; Ozuem et al., 2023). Interviews were recorded with participants' consent and transcribed verbatim. Following the systematic qualitative approach (Gioia et al., 2013), three researchers then read and analyzed the transcripts using commitment-trust theory and the personalization-privacy paradox as analytical lenses. The collaborative analysis followed a structured, multi-phase process consistent with recommendations for team-based qualitative analysis (Cornish et al., 2014; Richards and Hemphill, 2018). The team held weekly meetings throughout the cleaning and coding phases, beginning with independent proofreading of verbatim Zoom transcripts to remove personal identifiers and filler words, then convening to agree on the cleaned dataset. Over a two-month period, each researcher independently read through the transcripts and highlighted significant codes relating to trust and the personalization-privacy paradox. The team then engaged in consensus coding (Richards and Hemphill, 2018): researchers met to compare their independently identified codes, discussed discrepancies, and reconciled differences through dialogue until agreement was reached on which quotes belonged to which emerging themes. This process ensured that the researchers' analysis aligned with the participants' expressed perspectives, thereby enhancing the credibility of the insights (Guba and Lincoln, 2003; Richards and Hemphill, 2018). To further enhance the study's trustworthiness, the four researchers' diverse disciplinary backgrounds and national contexts facilitated investigator triangulation (Denzin, 2017; Ozuem et al., 2024; Patton, 2015; Stalmeijer et al., 2024), allowing the findings to be interpreted through multiple analytical and cultural lenses.

The coding then proceeded through three analytical stages. In the first stage, researchers retained only primary codes that appeared consistently across participants. In the second-order stage, researchers conducted inductive analysis to develop themes with implicit meanings by interpreting the primary codes directly taken from participants' responses (Gioia et al., 2013; Ozuem et al., 2022, 2024). The team convened to discuss and compare these emerging themes. In the third-order stage, the first-order and second-order codes were systematically organized into four theoretically distinct themes connected to existing concepts identified in the e-commerce AI literature. Throughout the process, all codes, quotes, and thematic assignments were organized in a shared spreadsheet stored on a password-protected university cloud platform, serving as a transparent audit trail. No dedicated qualitative analysis software was used; the research team relied on Google Docs for transcript cleaning and shared spreadsheets for code organization to maintain accessibility across team members.

The transition from the four thematic mechanisms to the final pathway framework (UCCG) occurred through a subsequent analytical step. The research team examined how the mechanisms co-occurred across individual participants, identifying patterned combinations; for example, participants who exhibited strong affective orientations toward convenience alongside low stereotypical cognition and minimal trust aversion represented a distinct cluster compared to participants who exhibited high trust aversion and strong demands for governance.

These patterned combinations were then mapped onto commitment-trust theory and the personalization-privacy paradox to develop the four consumer pathways described in the *Discussion* section. All codes and themes were constantly discussed and revised among researchers through this iterative process, and the final thematic structure was reviewed and approved by all members of the research team before the pathways were formalized.

5. Interpretation of themes

The analysis revealed four interlocking mechanisms that shaped how consumers navigated trust and privacy in AI-enabled e-commerce. This section provides a concise description of these mechanisms, followed by a detailed examination of each one. First, consumers' affective orientations capture enduring emotional dispositions, which is how AI makes consumers feel as humans, users, and data subjects. Second, consumers' stereotypical cognition illustrates pre-existing, generalized assumptions about AI's inherent limitations, comprising what consumers believe about AI capabilities before any specific brand interaction. The third mechanism, trust aversion, captures a firm-directed attribution pattern in which consumers perceive AI as serving retailer interests over their own, explaining why consumers withdraw trust in response to specific experiences of intrusive targeting or opaque data practices. The key distinction between stereotypical cognition and trust aversion is their object and trigger: stereotypical cognition is directed at AI as a technology and is carried into interactions as a prior belief, whereas trust aversion is directed at the firm deploying the AI and is activated by specific experiences that signal misaligned interests. The last mechanism, trust calibration, demonstrates consumers' expectation that trust must be actively earned through transparency and user control, indicating how trust could be restored or maintained. While these mechanisms are analytically distinct, they interact in patterned ways that give rise to the consumer pathways discussed in Section 6 *Four Consumer Pathways for AI in e-commerce (UCCG)*. Table 3 presents the definitions and keywords selected from the interviews for each mechanism.

Table 3
Thematic categories.

Themes	Definitions	Keywords
Consumers' affective orientations	Relatively stable, emotionally inflected dispositions that shape how consumers interpret the presence of AI in e-commerce. These orientations encompass feelings, intuitive comfort or discomfort, and rapid judgments about trust, control, utilities, and perceived human-likeness, which together influence whether consumers accept, tolerate, or resist AI. They concern not only evaluations of what AI can do, but also how AI makes consumers feel as users, as data subjects, and as human beings in e-commerce environments.	Accessibility Timesaving Decision-making Personalization Privacy concerns Monitoring Tracking purchases Being watched Violating Inauthentic Algorithmic Uneasy Creepy
Consumers' stereotypical cognition	A tendency to approach AI tools with initial suspicion rather than trust. It is grounded in three linked beliefs: that AI remains immature and error prone, that human judgment is more credible and should validate AI outputs, and that AI systems may introduce bias in selecting, summarizing, or promoting information. Consequently, consumers confine AI to low-stakes tasks and seek evidence, transparency, or human involvement before allowing it to influence their purchase decisions.	Waste of time Frustrated Experimental Dangerous Mistakes Human reviews Expertise Search/check for information Verify Biases Hallucinations Scams
Consumers' trust aversion	A tendency to assume that AI recommendations prioritize the retailer's interests over their own. It is activated when targeting appears intrusive or excessively personalized, or when data practices lack transparency, prompting trust to shift from the AI to the firm deploying it. In this state, consumers withhold trust, seek independent verification, and show reduced willingness to follow AI-generated suggestions.	(Not) trustworthy Wary Skeptical Uneasy Turned off Uncomfortable Being tracked Double-checking Don't fully trust (Targeted) AI advertising
Consumers' trust calibration	The belief that trust in AI shopping tools must be actively earned and appropriately adjusted through transparency and user control. When firms clearly disclose that AI is used, specify the data it relies on, and explain how that data is protected, consumers perceive the system as open and accountable. When users can opt in, disable features, or restrict data use, they experience their autonomy as respected.	More regulation Explaining the algorithm Full transparency Inform customers Privacy notice Disclaimer Should not be the default Data protection Not invasive Choice to choose

5.1. Consumers' affective orientations

Consumers' affective orientations refer to relatively stable, emotionally inflected dispositions that influence how consumers interpret the presence of AI in e-commerce. These orientations encompass feelings of comfort or discomfort, intuitive judgments about trust and control, and emotional responses to AI's perceived human-likeness. These factors collectively determine whether consumers accept, tolerate, or resist AI (Pelau et al., 2021; Poushneh, 2021). Participants expressed certain enduring feelings that guided whether they would lean in, set boundaries, or push back, even when the AI functional benefits were clear to them. Specifically, they indicated how they juggled their desire for cognitive relief against anxieties about surveillance, privacy, authenticity, and the erosion of human agency. This complexity was shown in four different categories. The first category shared by participants was cognitive ease provided by AI through personalization and information suggestions. For example, participant 10 shared their online-shopping experience with AI usage:

> I would say hyper-personalization. It's easier… You wouldn't find the whole availability of stuff you wanted, and AI-powered shopping made it easier to just type in what you want and actually find more specific products.

Participant 16 also said:

> I think personalization is the best thing. For ChatGPT, for me, the more I tell it, the more it's specific to me, and the more I want to use it. If it's more personalized, then people definitely want to use it more.

When AI remembers preferences, narrows choices, and serves relevant options, it reduces effort and feels efficient. Through targeted information, participants had a more helpful and time-saving experience with AI, which made AI play an essential role in their decision-making process. As participant 5 pointed out:

> I do find them helpful because I get to see, out of thousands of reviews that a product can have, it summarizes it into a couple of bullet points that make it easier to read.

Similarly, participant 15 stated:

> … The information gathering power that AI tools have is very crucial to the decision-making of purchase.

The above comments demonstrated the perceived value of AI in e-commerce in terms of information processing and reduced cognitive overload. However, in the second identified category, some participants showed mixed or ambivalent feelings about the risks of privacy while having the personalized experience provided by AI, aligning with the personalization-privacy paradox (Ameen et al., 2022; Cloarec, 2020). With the use of AI in e-commerce, participants enjoyed the convenience but ran a constant emotional trade-off in the background, aligning with the personalization-privacy paradox. Participant 5 explained:

> We want privacy, but at the same time it's giving me exactly what I want, and you can't do that without getting the information from the customer. So, you're kind of between a rock and a hard place.

This statement is also supported by participant 6:

> … There's a part of me that wants to keep using it and doesn't really care, but a bigger part of me thinks that the information is going to the government or those theories people talk about. I feel uncomfortable about how much it knows, and when you log back into it as well, when it saves previous chats.

Besides the second category related to the personalization-privacy paradox, when AI responses felt formulaic, participants perceived the experience as artificial and distant, which falls under the third category. This created an apathetic or negative affective orientation among participants, because the system was not seen as responsive to nuance or context. Participants 13 and 31 expressed that as chatbots' answers tended to be very automated, they swayed consumers from talking to them, raising the need for these tools to become less artificial. For example, participant 1 pointed out:

> In speaking to authenticity, it does feel like a very inauthentic and inorganic way of presenting products. Right? It does feel very algorithmic.

For others, hyper friendliness or faux empathy triggered suspicion about the source and truth of the information. Specifically, participants expressed their emotional responses to the AI's attempts to emulate human behavior during their shopping experiences. In these cases, human mimicry lowers trust instead of raising it and makes consumers feel uneasy (Participant 4). Participant 28 said:

> Whenever it tries not to be super-human… weirdly, I don't like that. If it starts adding a bunch of words that's trying to sound friendly, I'd rather it stays robotic. Because whenever they're adding all those words, it makes me question the validity of the information.

In the last category, a subset of participants framed AI in civilizational or human-development terms. Participants 3, 8, and 12 used terms such as "creepy" and "scary" to describe the fact that AI predicted their shopping preferences and offered them personalized recommendations. Participant 25 also expressed their concern:

> I hope that as a society, we can continue to keep remembering that, because when we have something so precious… our style… that gives people insight into who we are. It gives insight into our individuality and our personability, and to let AI strip that away from us, that's very sad. That's a sad world to think about.

Participants' comments indicated that they worried predictive systems may weaken human agency, problem solving, or adaptability. This produces a precautionary or even defensive disposition, in which individuals still want to retain space for unpredictability, human error, and learning.

5.2. Consumers' stereotypical cognition

Consumers' stereotypical cognition refers to a predisposition to approach AI tools with initial skepticism rather than trust. This phenomenon is underpinned by three interconnected beliefs: that AI remains immature and prone to errors, that human judgment is more reliable and should serve as a validation mechanism for AI outputs, and that AI systems may introduce bias in the selection or promotion of information. This construct reflects what consumers assume about AI in general (Yamamoto and Suzuki, 2025) before engaging with any AI-enabled e-commerce platform, and it leads them to confine AI to low-stakes tasks and seek evidence or human confirmation before allowing AI to influence purchase decisions. Participants showed that they approached AI tools with various ready-made assumptions. This cognition was shaped, first, by perceptions of technological immaturity, for example, the belief that AI chatbots misunderstood intent, provided partial or hallucinated answers, and were released as finished products even though they were still being tested. Participant 22 explained:

> I don't ever use a chatbot, because they never work, and they're always a waste of time, but the summaries of the reviews are helpful.

Participant 4 also shared:

> It is still something that is in the works, so it doesn't always make the best recommendations.

To some participants, AI lacked the capabilities to fully understand what they needed, making AI seem prone to mistakes and leading them to dislike it (Participants 5 and 11). This notion is reinforced by an anthropocentric bias, meaning that individuals give greater credibility to human judgment, lived experience, and identifiable reviewers. For instance, participants often cross-checked AI suggestions with Google, friends or experts, because human input was seen as richer, more accountable, and more context sensitive. Participant 10 noted:

> I still feel it's very important for me, for a lot of customers, that AI is still not substituting human suggestions… I still need to look for human opinions, even more than before… now it's even harder to find information suggesting… someone actually uses something, so I still feel like, no matter how good they get, we will still need human input… you still need to search for human opinion.

Similarly, participant 3 stated:

> I really base my shopping on other people's reviews, like what real people think in their experiences with products. So, I can see it becoming a tool that I use, but not like my main source of recommendations or decision making.

The need for human judgement might have come from awareness of algorithmic bias, that is, the sense that AI can over select positive content, ignore dissenting reviews, rely on a narrow set of data sources, or be manipulated to promote certain products, which creates doubts about fairness and accuracy (Bose, 2024; Lambrecht and Tucker, 2019; Manyika et al., 2019; Obermeyer et al., 2019). Participant 6 mentioned:

> … AI can be great, but when it's feeding you information that's untrue or targeted against you, then you're just being fooled by it… It just gives you incorrect information that ends up leading to you buying something or purchasing something that doesn't help you… or that could even not exist.

This phenomenon where consumers have a hard time accepting AI decisions is referred to as algorithm aversion (Yamamoto and Suzuki, 2025). However, this aversion highly depends on context, and acceptance of AI is not simple 'people like AI' or 'people dislike AI' (Yamamoto and Suzuki, 2025). In this situation, participants needed AI-powered tools to show them proof or citations for the provided information to ensure they did not fall for scams (Participants 7, 8, and 18). Specifically, participant 8 stated how they would not trust AI and handle the information verification by themselves:

> I think if it shared… the resources, the original links… If I could be able to verify everything that it's telling me, then I would probably trust it more… I always double-checked. I never just trusted the AI.

The above quotes illustrated a cognitive shortcut, in which consumers started from suspicion rather than from trust, limited their reliance on AI to low stakes tasks, and required evidence or human confirmation before allowing AI to influence their e-commerce decisions. As previously discussed, while both stereotypical cognition and trust aversion manifest skepticism toward AI, they differ in their target and trigger. Stereotypical cognition targets AI as a technology and is illustrated in interactions as a prior belief, reflecting consumers' general assumptions about AI before engaging with any specific platform. Trust aversion, by contrast, focuses on the firm employing AI and is activated by experiences, such as intrusive retargeting or hyper-personalized advertisements, that indicate misaligned corporate interests. The next section elaborates on this firm-directed dimension.

5.3. Consumers' trust aversion

Consumers' trust aversion refers to a tendency to assume that AI recommendations prioritize the retailer's interests over the consumer's own. Unlike stereotypical cognition, which is directed at AI as a technology and reflects prior beliefs, trust aversion is directed at the firm deploying the AI and is activated by specific experiences (e.g., retargeting across platforms or hyper-personalized advertisements) that signal misaligned interests between the consumer and the retailer. Consistent with prior research showing that covert personalization evokes vulnerability and diminishes engagement (Aguirre et al., 2015), consumers in this state withhold trust, seek independent verification, and show reduced willingness to follow AI-generated suggestions. Participants' negative presumptions of AI tools might be worsened if they perceived that AI was using their personal data to steer them toward outcomes that benefited corporations more than themselves, or when AI-enabled product promotions felt intrusive or too perfectly targeted, which created a feeling of being watched and managed. For instance, participant 18 stated:

> And for big companies like that, I would say it's a little uneasy when I feel a lot of people use AI for [advertising], because it seems like they're taking a short route.

Participant 29 also said:

> … I'll see an AI ad of something I already bought, or close to something I already bought, and it makes me feel uncomfortable… And I feel like if they did that less, I would trust them more.

Many prominent companies, such as Netflix, Amazon, and Adidas, have leveraged AI to enhance the quality of their marketing campaigns (Gunasekar et al., 2023; Mathur, 2023; Prentice et al., 2020; Sevaslidou et al., 2024). Some AI-generated advertisements were deemed to be more effective than conventional ones (Nesterenko and Olefirenko, 2023). However, AI promotions in e-commerce do not always have positive impacts. In certain instances, AI-generated images heightened users' resistance to AI due to consumers' perceived threat (Mou et al., 2024). Participant 11 clarified:

> What would make me uneasy is if I started getting targeted ads on other platforms; that would probably be, oh, my information is being spread elsewhere.

Once again, this situation is traced back to the personalization-privacy paradox mentioned above where consumers would use AI for personalized recommendations but would not fully trust them (Ameen et al., 2022; Cloarec, 2020). Participants mentioned that they would compensate for this issue by double-checking information or seeking human-made information before acting. This trust aversion does not always mean outright rejection of AI, but from our interviewees' responses, it does lower immediate compliance with AI recommendations. This could erode brand trust over time, especially when targeting feels excessive or when previous negative or second-hand experiences make the system seem risky or easily abused.

5.4. Consumers' trust calibration

Trust calibration refers to the consumer-driven belief that trust in AI shopping tools must be actively earned and appropriately adjusted through transparency and user control. When companies clearly disclose that AI is utilized, specify the data it relies on, and explain how that data is safeguarded, consumers perceive the system as transparent and accountable. Additionally, when users have the option to opt in, disable features, or restrict data usage, they experience their autonomy as respected, which resonates with existing research on AI personalization and privacy (Qadri et al., 2026). In the interviews, participants described that trust in AI-enabled e-commerce could be increased if companies clearly demonstrated that AI was used, how AI worked, what consumer data was leveraged, how long the data was kept, and their efforts in developing safeguards and regulations around the AI development. Participants would trust AI more in e-commerce if the companies explained how AI recommendations were made. For example, participant 20 noted:

> … Just saying if it is AI, and how they're using AI, just so I'm aware how it's being used, and… how they're pulling my data from… where the data is coming from that they're making suggestions from.

Participants 5, 16, and 28 shared the same perspectives that transparency in plain language, placed where people will see it (e.g., short labels such as "AI-generated" or "AI recommended," or brief notices about data sources and storage), signaled that the company was not hiding the role of AI. This outcome concurs with research indicating that to deal with privacy issues, corporations need to provide transparent labeling and data usage disclosures, particularly offering users the option to select which information to disclose (Aguirre et al., 2015; Ma et al., 2026; Qadri et al., 2026).

Participant 28 shared:

> It should have a lot of transparency… I don't really get the sense of hiding how your AI functions… I guess there's a competitive advantage… but at the same time, your competitors are already gonna have it anyways… 90% of the time that companies are transparent with their customers on their methods, they end up being way more successful anyways.

Similarly, participant 16 shared:

> I would love full transparency… It would establish more trust between the consumer, and it would make them feel more at ease with knowing AI is involved.

However, transparency or privacy features in some cases only worked for willing individuals, and not those who were unwilling to be profiled (Awad and Krishnan, 2006). The combination of transparency and user control could change this narrative. Specifically, participants in this study linked trust enhancement to user control. They wanted to be able to turn AI features on or off, to opt in rather than face AI by default, to choose what data can be used, and to deny tracking when they prefer privacy. Participant 15 suggested:

> I would say definitely have two versions of the platform, so it should have a certain button, or a certain sort of way to turn on or turn off the AI tools to help you on the online purchase.

On the same note, participant 7 stated:

> I think giving people the choice to choose, and being transparent with them that way, and say, like "Hey, this is what we're gonna use it for," rather than just kind of opting them in by default, which I know is usually to the benefit of the company.

When participants could choose the intensity of AI involvement, they felt that their autonomy was respected, which in turn supported trust. The notion of having humans in the loop does not only increase trust but also provides humans the ability to judge and stay accountable for their

usage (Te'eni et al., 2025). Taken together, trust calibration in this context is the consumer-driven idea that trust is not automatic. It is earned through visible transparency and through controls that let users match AI assistance to their own comfort level.

6. Discussion

The results indicate that trust in AI-enabled online shopping is not a binary state, but a continually adjusted, experience-based judgment shaped by AI performance, communicative style, and data practices encountered during the shopping journey. Using an informed thematic approach (Gioia et al., 2013), this study moved from informant-centric first-order codes (e.g., "time-saving," "creepy," "I double-check," "too targeted," "opt-in not default") to second-order themes that captured latent mechanisms, and then consolidated these into the four aggregate dimensions detailed in Section 5 *Interpretation of themes*: affective orientations, stereotypical cognition, trust aversion, and trust calibration.

Building on these themes and grounded in the commitment-trust theory (Morgan and Hunt, 1994) and personalization-privacy paradox (Cloarec, 2020), the researchers developed an integrative pathway framework that explained how patterned combinations of these mechanisms generated distinct routes to commitment, reuse, and loyalty in AI-driven brand experiences: a utility-led pathway (U) where instrumental benefits dominate and privacy tensions remain backgrounded; a control-led pathway (C) where personalization value is contingent on visible governance, opt-in choice, and explainability; a communication-led pathway (C) where perceived authenticity and tone determine whether trust can form despite functional utility; and a governance-led pathway (G) where agency protection and perceived opportunism constrain reliance unless strong safeguards are present. Across the pathways, consumers are not uniformly trusting or privacy-indifferent. Instead, trust and perceived privacy risks are calibrated in patterned ways that shape commitment and loyalty to AI-enabled platforms. The pathways, therefore, clarify when trust becomes a basis for relationship commitment and when privacy concerns or perceived opportunism interrupt that commitment-trust link (see Fig. 1).

6.1. Four consumer pathways for AI in e-commerce (UCCG)

6.1.1. Utility-led trust (U)

The first pathway, utility-led trust, captures consumers who approach AI-enabled platforms with affective orientations that center on timesaving, accessibility, decision-making support, and helpful personalization. Previous studies showed that AI trained on extensive datasets effectively enabled hyper-personalization and enhanced user engagement on e-commerce platforms (Lv and Huang, 2024; Madanchian, 2024). From a commitment-trust perspective, this early confidence in the AI's functional capabilities and its good intentions (Yamamoto and Suzuki, 2025) increases commitment to the platform, encouraging cooperative behaviors and reuse.

The personalization-privacy paradox remains low in salience because the perceived value of tailored experiences clearly outweighs perceived privacy costs (Canhoto et al., 2024; Lo et al., 2026), so data exchange feels 'worth it' and privacy concerns are backgrounded. Stereotypical cognition around AI bias or hallucination is weak, with consumers occasionally verifying outputs but generally defaulting to acceptance, which keeps the trust aversion low. In this pathway, discomfort emerges only when targeting messages feel 'too perfect' or 'creepy' although perceived opportunism remains weak, so the chain from trust to commitment and loyalty stays intact. For these consumers, light-touch transparency, such as straightforward privacy notices and visible data protection practices, is typically sufficient to calibrate trust. Opt-in or opt-out controls are viewed as nice enhancements rather than essential safeguards. As a result, trust signals remain strong, personalization-privacy paradox tensions are minimal, and the outcome is rapid reuse and sustained loyalty to the AI-enabled platform.

6.1.2. Control-led trust (C)

In the second pathway shaped by privacy tension, consumers experience mixed feelings toward AI-enabled personalization: they recognize its usefulness but feel uneasy and watched, so the personalization-privacy tradeoff is very salient. From a commitment-trust lens, this affective ambivalence keeps trust from reaching a level that would support strong commitment to the platform. Consumers want relevance without surveillance, which reflects a classic personalization-privacy paradox

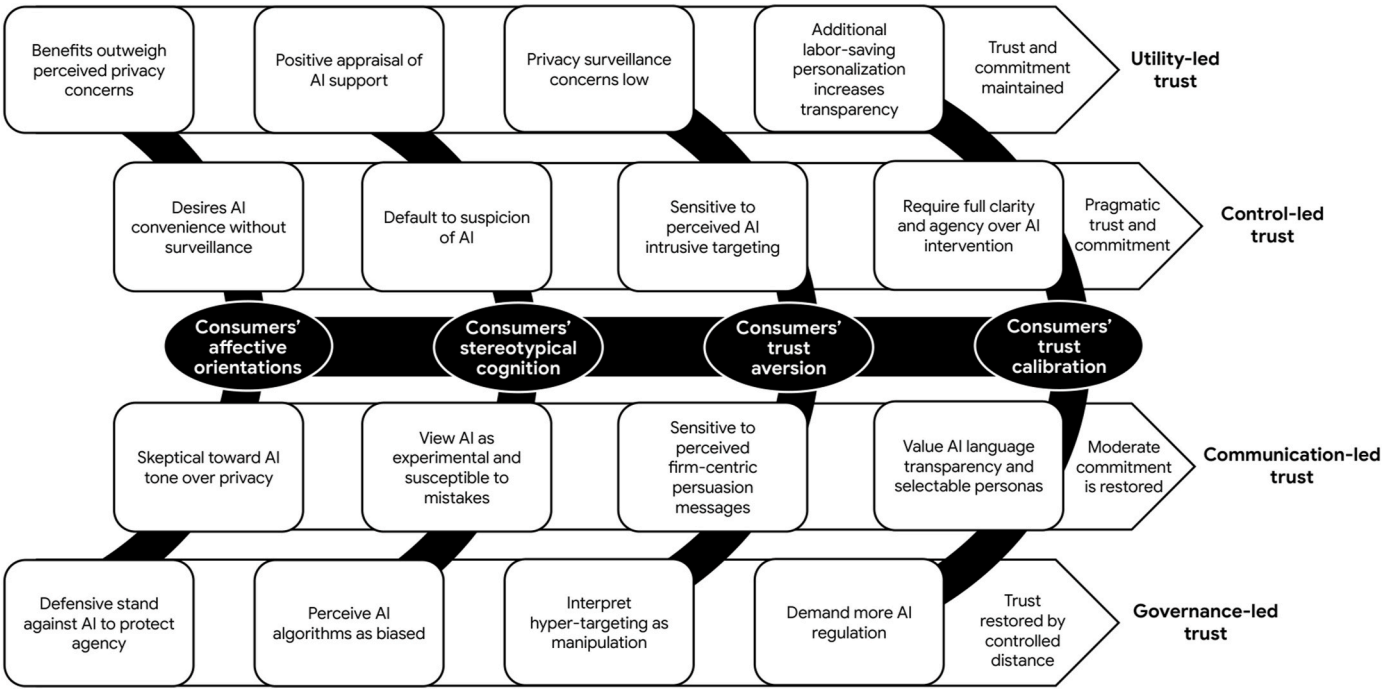


Fig. 1. Consumer pathways for AI transformation in e-commerce.

(Cloarec, 2020). Cognitively, their default stance is cautious; they actively verify AI outputs, seek citations and human reviews, and treat trust as conditional rather than given, keeping commitment bounded and easily reversible. Targeted AI ads and retargeting can trigger trust aversion, which results in weaker loyalty and greater switching propensity along the commitment-trust chain.

To calibrate trust, these consumers require robust control: full transparency about data use and algorithms, clear explanations, and explicit opt-in rather than passive default enrollment. When platforms provide such controls, consumers' sense of agency is gradually restored. Hence, trust is raised to a calibrated, workable level, and commitment becomes pragmatic and limited rather than deep. Under these conditions, the personalization-privacy paradox becomes more manageable because personalized modes accessed via explicit opt-in can reduce tension and support ongoing, but guarded, engagement.

6.1.3. Communication-led trust (C)

In this pathway, consumers responded to AI primarily based on tone and style rather than functional benefits. Engaging or warmly anthropomorphic AI interactions can foster trust (Calhoun et al., 2019; Ding and Najaf, 2024; Poushneh, 2021). However, communication that feels inauthentic, overly algorithmic, or 'too human' in a faux empathetic way produces resistance (Ciechanowski et al., 2019; Y. Li et al., 2023; Pelau et al., 2021), which threatens perceived integrity and shared values within a commitment-trust perspective and prevents commitment from forming. For these consumers, the personalization-privacy paradox is secondary; communication style is the primary trigger. Cognitively, they would view AI as experimental and prone to errors or hallucinations due to the lack of representativeness and biases in the data used to train AI models (Madanchian, 2024; Singh and Mishra, 2024), favoring human expertise and cross-checking. They would only build trust after seeing consistent, modest performance over time.

For communication-led consumers, trust aversion is easily activated when messages feel manipulative or persuasion heavy, which they experience as firm-centric opportunism, thereby stalling commitment along the commitment-trust chain. To calibrate trust, consumers prioritize plain-language transparency and the ability to select AI personas, ranging from strictly robotic to warm and approachable. They also value granular control over e-commerce AI behaviors (e.g., empathy, fun, or spontaneity) to ensure the tool meets their needs. This resulting persona fit functions as a trust antecedent that increases perceived alignment and allows for moderated commitment. Even then, transparent data practices and privacy guardrails remain necessary to manage the personalization-privacy paradox, but they are not sufficient without an authentic and well-calibrated communication tone.

6.1.4. Governance-led trust (G)

In the fourth pathway, consumers adopt a defensive stance, often experiencing AI as creepy or scary and focusing on preserving individuality and human agency by confining AI to low-stakes, low-involvement tasks. For them, the personalization-privacy paradox is fundamentally agency-reducing, so baseline trust is low, and commitment is deliberately restrained within a commitment-trust perspective. Cognitively, they exhibit a strong anthropocentric bias that places human judgment above machine output, maintain high awareness of algorithmic bias, and insist on verification, citations, and human reviews, extending trust only when visible human oversight is present.

Additionally, hyper-targeted personalization is read as manipulation; for example, when personalization feels intrusive, unfair, or opaque, it can diminish the positive impact of trust on satisfaction (Aguirre et al., 2015), which heightens trust aversion, reduces willingness to share data, and supports preferences for manual search; perceived opportunism effectively severs the link between trust and deep commitment. These consumers demand stronger governance to calibrate trust: more regulation, robust data protection, explicit opt-in, granular controls, and non-invasive defaults. This is consistent with the literature that anticipates a heightened need for accountability and responsible AI practices from businesses (Barat and Gulati, 2024; Qadri et al., 2026; Rakova et al., 2021). Under such strict governance conditions, they may accept limited personalization and maintain a controlled distance that enables coexistence with AI, but this rarely translates into strong loyalty or relational commitment.

Overall, the four pathways reveal a structural pattern in how trust and the personalization-privacy paradox interact. Where trust forms easily (utility-led), the paradox is backgrounded; where trust formation is conditional (control-led, communication-led), the paradox becomes salient as a calibration mechanism; and where trust formation is constrained (governance-led), the paradox is interpreted as fundamentally agency-reducing. This patterned variation suggests that trust and privacy are not separate determinants of AI adoption but jointly constitutive of how consumers relate to AI-enabled commerce. More importantly, these pathways are not fixed consumer types but orientations that could shift in response to changing experiences. A utility-led consumer who encounters a data breach, a 'creepy' retargeting experience, or a publicized AI failure may shift toward a governance-led orientation as trust aversion is activated, and the demands for safeguards intensify. Conversely, a governance-led consumer who encounters a platform with strong transparency features, explicit opt-in controls, and clear algorithmic explanations may shift toward a control-led engagement as their calibration requirements are met. Similarly, a communication-led consumer who experiences consistent AI interactions over time may gradually move toward utility-led trust as functional confidence builds. Understanding these transition conditions is essential as they suggest that retailers can influence which pathway consumers adopt by adjusting their transparency, control, and communication design choices, which is further addressed in Section 6.2.2 *Managerial contribution.*

6.2. Contributions of the study

6.2.1. Theoretical contribution

Building on relationship marketing and human-AI trust research, this study extended commitment-trust theory (Morgan and Hunt, 1994) and the personalization-privacy paradox (Awad and Krishnan, 2006; Cloarec, 2020) in two ways that address specific limitations in each framework. First, commitment-trust theory has been productively applied to explain relational outcomes in e-commerce and AI contexts (Elbeltagi and Agag, 2016; Ghali-Zinoubi, 2023; Safari, 2026; Shrivastava, 2025; Willis et al., 2021), but it treats trust as a relatively stable variable of commitment rather than as a dynamically calibrated process. This is a significant limitation in AI-enabled online shopping, where trust can shift rapidly in response to disclosure effects, communication style, or perceived data misuse (Gu et al., 2024; Luo et al., 2019). The present study addresses this limitation by theorizing trust in e-commerce AI as a continuously adjusted, experience-based judgment (i.e., trust calibration) rather than a fixed input to the commitment-trust chain. The UCCG framework specifies when and for whom trust leads to commitment (e.g., utility-led consumers) and when perceived opportunism or privacy concern interrupts that link (e.g., governance-led consumers).

Second, the personalization-privacy paradox has been applied across multiple retailing contexts (Aguirre et al., 2015; Ameen et al., 2022; Canhoto et al., 2024), but prior applications have largely treated the paradox as a monolithic cost-benefit calculation that operates uniformly across consumers. This study advances the framework by demonstrating that consumers navigate the paradox through systematically different orientations. Some consumers prioritize functional utility over privacy concerns; others demand robust governance, and still others prioritize communication authenticity as the primary filter. The UCCG pathways translate these differences into a conceptual architecture that explains how patterned combinations of trust and privacy perceptions generate

distinct routes to commitment, reuse, and loyalty. Conceptually, the study also advanced the literature by identifying four interlocking mechanisms (i.e., affective orientations, stereotypical cognition, trust aversion, and trust calibration) that capture distinct aspects of how consumers process e-commerce AI experiences. In doing so, it helps reconcile mixed prior evidence on privacy disclosure and trust (Aguirre et al., 2015; Ameen et al., 2022; Gupta, 2025) by showing that disclosure, tone, and perceived opportunism become decisive at different points along the pathway, depending on the consumer's dominant orientation.

6.2.2. Managerial contribution

The findings offered actionable guidance for designing and governing AI-enabled shopping experiences around trust calibration rather than one-size-fits-all 'more personalization' logics. The implications discussed below are drawn from the patterns observed across this qualitative dataset and should be read as theoretically grounded suggestions for retailer practice, illustrative of how the UCCG pathway logic could inform trust-centered AI design, rather than as universally applicable prescriptions. For utility-led consumers, the findings suggest that light-touch transparency may be enough; over-engineering transparency for this segment risks disrupting the experience that underpins their trust and loyalty (Aguirre et al., 2015). Concrete applications include clear 'AI-generated' labels on product review summaries, straightforward privacy notices, and visible data protection badges. For control-led consumers, the interviews indicate that meaningful user control is particularly valued. Retailers may consider implementing explicit opt-in rather than passive default enrollment for AI-powered personalization, provide toggles that allow consumers to disable specific AI features (e.g., retargeting, cross-platform recommendations), and offer granular data permissions that let consumers choose what information is used and for how long. These controls may reduce privacy tension and perceived opportunism (Awad and Krishnan, 2006; Canhoto et al., 2024; Qadri et al., 2026). For example, e-commerce AI platforms could offer a 'personalization settings' dashboard where consumers can see exactly which data points are used for recommendations and selectively opt out.

For communication-led consumers, the interviews suggest that AI interaction style is a particularly important factor. Retailers may benefit from avoiding over-anthropomorphized features that can trigger uncanny-valley effects and authenticity concerns (Ciechanowski et al., 2019), while ensuring AI chatbots maintain a consistent, authentic tone. Offering persona-selection features, where consumers can choose among various conversational styles (e.g., robotic or warm tones), could allow this segment to control the interaction tone and build trust over time (Igarashi et al., 2026; M. Li and Wang, 2023; Lu and Zhang, 2025). For governance-led consumers, the findings suggest that strong regulatory signals and human oversight may be important to sustain engagement. Retailers may benefit from prominently displaying compliance with data protection regulations (e.g., the EU AI Act or the California Consumer Privacy Act; California Legislative Information, 2018; EU Artificial Intelligence Act, n.d.), provide granular controls and non-invasive defaults, and ensure that human consumer service remains accessible as an alternative to AI-driven interactions. This segment is most sensitive to perceived manipulation, so transparency may need to extend beyond labels to include clear explanations of how algorithmic recommendations are generated and what safeguards prevent bias (Barat and Gulati, 2024; Kamoonpuri and Sengar, 2023; Rakova et al., 2021).

Of broader significance, the interviews revealed a widespread lack of awareness among participants regarding AI features or integration in their e-commerce experiences. Many participants, including those who had interacted with AI in e-commerce settings (participants 1, 6, 10, and 18) and noticed features like Amazon's Rufus or AI review summaries, were initially unsure whether their experiences involved AI. This observation underscores the absence of transparency regarding AI integration practices between firms and consumers and reinforces the finding that trust calibration cannot occur when consumers are unaware that AI is being used. E-commerce platforms may benefit from more transparently disclosing AI integration to support trust calibration (Teodorescu et al., 2023) and helping consumers recognize how AI is embedded in their online shopping experiences.

6.3. Limitations and future research

While insightful, the results are constrained by the sample's demographic profile; specifically, the participant pool skewed heavily toward young adults (18-24) in the U.S. Although this composition is theoretically justified—the demographic represents the heaviest users of AI-enabled shopping tools and thus constitutes an information-rich population for exploring the personalization-privacy paradox (Commerce, 2025; EMARKETER Editors, 2025; PayPal, 2025), and research on the behaviors of younger consumers to investigate AI-related phenomena frequently indicates emerging patterns (Bunea et al., 2024; Guo and Luo, 2023)—it limits the direct transferability of the findings to older consumer cohorts who may bring different baseline orientations toward AI, distinct privacy expectations, or lower familiarity with AI-driven interfaces. In addition to the age concentration, the sample is also geographically concentrated, with most participants residing in the Northeast region and a smaller number drawn from other regions in the U.S. This regional concentration further constrains transferability and suggests that future research should target geographically dispersed samples to test whether the UCCG pathways hold across different regional contexts and digital retail environments.

Additionally, many participants were initially unaware that they had been interacting with AI in their e-commerce experiences, and the self-reported nature of their AI usage and trust perceptions introduces retrospective interpretation that may not fully capture in-the-moment responses. The data also reflected a particular moment in the rapid evolution of AI tools; as AI becomes more ubiquitous, consumers' affective orientations, trust calibrations, and privacy expectations are likely to shift. Future research could test the UCCG pathways quantitatively at scale to assess their prevalence and predictive validity across broader and more age-diverse consumer populations, including consumers over 50 whose AI adoption patterns and trust dynamics may differ substantially. Longitudinal designs would be crucial to examine how consumers move between pathways over time in response to changing AI features, policy interventions, or data breaches, while cross-cultural research could test whether the mechanisms identified here hold in non-Western e-commerce AI contexts, where baseline privacy expectations and trust norms may differ. Investigating vulnerable or marginalized consumer groups, as well as high-stakes contexts where AI decisions have significant consequences, would further deepen understanding of how trust, personalization, and privacy interact in AI-driven brand experiences. Additionally, as AI-enabled retail evolves toward agentic AI and fully autonomous shopping agents, examining how these developments reshape trust calibration dynamics would extend the framework into the next generation of retail experiences.

7. Conclusion

This study shows that trust and the personalization-privacy paradox are not separate determinants that consumers weigh independently when adopting AI in e-commerce. Instead, consumers actively calibrate trust in e-commerce AI as they navigate the tension between personalization and privacy. Different consumers do not simply trust AI more or less; they do fundamentally different things with AI—leveraging it for personalization, negotiating its terms, evaluating its style, or containing its reach. The results open a new discussion on what 'AI adoption' means in future retailing scholarship and practice: it may not be a single threshold consumers cross but a calibrated orientation that shapes the kind of relationship a consumer is willing to have with an AI-enabled platform.

This study's contribution sits at the intersection of these insights. By integrating commitment-trust theory with the personalization-privacy paradox, the UCCG framework offers retailers a qualitative, theoretically grounded vocabulary that prior research has documented but not theorized. The findings suggest that retailers' design choices around transparency, control, communication style, and governance may operate differently across consumer orientations, and that a single 'more personalization is better' logic is unlikely to serve all consumers equally. These suggestions reflect patterns observed within this qualitative sample and should be tested across broader and more demographically diverse populations. The integration of trust and privacy theorizing demonstrated here may extend to other algorithmically mediated consumer contexts (e.g., AI-driven financial services, healthcare consumer platforms) where relational outcomes are similarly entangled with data practices. As agentic AI and fully autonomous shopping agents become more prevalent, the pathways identified here may evolve in ways that current frameworks cannot anticipate, and longitudinal and quantitative replication will help track this evolution.

CRediT authorship contribution statement

Chi B. Vu: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing. **Wilson Ozuem:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Writing – review & editing. **Michelle Willis:** Formal analysis, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **James Katz:** Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jretconser.2026.104938.

Data availability

The data that has been used is confidential.

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